

Java class name: Controller Method signature: private void handle
addStudent()

Test Case #	Functional Requirement, or Test Objective	Test description and Input Data	Expected Result
1	This method should add a valid resident student.	- Enter First Name: Lucas \n- Enter Last Name: Barrales \n- Enter DOB: 2/24/2005 \n- Enter Major: CS \n- Enter Credits: 15 \n- Select Resident \n- Click Add Student	Student is added successfully and displayed in the output area.
2	This method should add a valid TriState student.	- Enter valid student data \n- Select TriState \n- Enter State: NY \n- Click Add Student	- Enter valid student data \n- Select TriState \n- Enter State: NY \n- Click Add Student
3	This method should add a valid international student	- Enter valid student data \n- Select International \n- Leave Study Abroad unchecked \n- Click Add Student	Student is added successfully and displayed as an International student
4	This method should reject a duplicate student.	- Add a valid student once \n- Re-enter the exact same First Name, Last Name, and DOB \n- Click Add Student again	Output displays error that the student already exists
5	This method should reject missing fields	- Leave one or more student fields blank \n- Click Add Student	Output displays error that all student fields must be filled.
6	This method should reject an invalid DOB format.	- Enter DOB as 2005-02-24 \n- Fill all other fields correctly \n- Click Add Student	- Enter DOB as 2005-02-24 \n- Fill all other fields correctly \n- Click Add Student
7	This method should reject an invalid	- Enter Major: BIO \n- Fill all other fields	Output displays error that the major is

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	major	correctly \n- Click Add Student	invalid.
8	This method should reject negative credits.	- Enter Credits: -3 \n- Fill all other fields correctly \n- Click Add Student	Output displays error that credits cannot be negative
9	This method should reject an invalid TriState state	- Enter valid student data \n- Select TriState \n- Enter State: NJ \n- Click Add Student	Output displays error that TriState state must be NY or CT.
10	This method should require a student type selection	- Enter valid student data \n- Do not select Resident, TriState, or International \n- Click Add Student	Output displays error to select a student type

Java class name: Controller Method signature: private void handleRemoveStudent()			
Test Case #	Functional Requirement, or Test Objective	Test description and Input Data	Expected Result
1	This method should remove an existing student	- Add a valid student first \n- Re-enter that student's First Name, Last Name, and DOB in the Students tab \n- Click Remove Student	Student is removed successfully and confirmation appears in output.
2	Student is removed successfully and confirmation appears in output.	- Enter a student that was never added \n- Click Remove Student	Output displays error: Student not found.
3	This method should reject removal when required identity	- Leave one or more of First Name, Last Name, or DOB blank	Output displays error: Student not found

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	fields are blank.	\n- Click Remove Student	

Java class name: Controller Method signature: private void handleEnroll()			
Test Case #	Functional Requirement, or Test Objective	Test description and Input Data	Expected Result
1	This method should enroll a valid student in a valid section.	- Add a valid student first \n- In Enrollment tab enter matching First Name, Last Name, DOB \n- Enter Course: CS100 \n- Enter Time: periodOne \n- Click Enroll	Output displays that the student was added to CS100 8:30.
2	This method should reject enrollment of a student not in the system	- In Enrollment tab enter a student that was never added \n- Enter valid course/time \n- Click Enroll	Output displays error: Student not found.
3	This method should reject a blank course.	- Enter valid enrollment student info \n- Leave Course blank \n- Enter valid time \n- Click Enroll	Output displays error: Course and time are required
4	This method should reject a blank time.	- Enter valid enrollment student info \n- Enter valid course \n- Leave Time blank \n- Click Enroll	Output displays error: Course and time are required.
5	This method should reject an invalid	- Enter valid enrollment student	Output displays error: Invalid course

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	course.	info \n- Enter Course: CS999 \n- Enter Time: periodOne \n- Click Enroll	
6	This method should reject an invalid time value	- Enter valid enrollment student info \n- Enter Course: CS100 \n- Enter Time: 10:30 \n- Click Enroll	Output displays error: Invalid time value.
7	This method should reject duplicate enrollment in the same course.	- Enroll a student in CS100 periodOne \n- Attempt to enroll the same student in CS100 periodOne again \n- Click Enroll	Output displays that the student is already enrolled in CS100
8	This method should reject a student who does not meet a major prerequisite	- Add a non-CS student \n- Try to enroll in CS300 periodThree \n- Click Enroll	Output displays a major prerequisite error.
9	This method should reject a student who does not meet a standing prerequisite	- Add a freshman student \n- Try to enroll in ECE400 periodSix \n- Click Enroll	Output displays a standing prerequisite error.
10	This method should reject a time conflict.	- Enroll a student in one section at periodOne \n- Try to enroll the same student in another section also at periodOne \n- Click Enroll	Output displays a time conflict error.
11	This method should reject a section that is full	- Enroll 4 students into the same section \n- Attempt to enroll a 5th student in that section \n- Click Enroll	Output displays that the section is full.

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12	This method should reject enrollment when the credit limit would be exceeded.	- Enroll a student until the current total is near 20 credits \n- Attempt to enroll one more course that would exceed 20 \n- Click Enroll	Output displays a credit limit exceeded error

Java class name: Controller Method signature: private void handleDrop()			
Test Case #	Functional Requirement, or Test Objective	Test description and Input Data	Expected Result
1	This method should drop a student from a valid enrolled section.	- Add and enroll a valid student in CS100 periodOne \n- In Enrollment tab enter matching student info, Course: CS100, Time: periodOne \n- Click Drop	Output displays that the student was dropped from CS100 8:30
2	This method should reject dropping a student not in the system.	- Enter a student that was never added \n- Enter valid course/time \n- Click Drop	Output displays error: Student not found.
3	This method should reject dropping from an invalid course.	- Enter valid enrollment student info \n- Enter Course: CS999 \n- Enter valid time \n- Click Drop	Output displays error: Invalid course
4	This method should reject dropping from an invalid time.	- Enter valid enrollment student info \n- Enter valid course \n- Enter Time: 10:30 \n- Click	Output displays error: Invalid time value.

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		Drop	
5	This method should reject dropping a section the student is not enrolled in.	- Add a student but do not enroll them in CS200 periodTwo \n- Enter Course: CS200 \n- Enter Time: periodTwo \n- Click Drop	Output displays that the student is not enrolled in this section.

Java class name: Controller Method signature: private void handlePrintStudents()			
Test Case #	Functional Requirement, or Test Objective	Test description and Input Data	Expected Result
1	This method should print all students in the system.	- Add multiple students \n- Click Print Students	Output displays all students currently stored in the system.
2	This method should show an empty-list message when no students exist.	- Start program with no students added \n- Click Print Students	Output displays: Student list is empty!

Java class name: Controller Method signature: private void handlePrintScheduleByCourse()			
Test Case #	Functional Requirement, or Test Objective	Test description and Input Data	Expected Result
1	This method should print all available sections ordered by course	- Click Print Schedule by Course	Output displays all initialized sections in course order.

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2	This method should include course, time, instructor, and classroom in each line	- Click Print Schedule by Course	Each printed section contains course number, meeting time, instructor, and classroom information

Java class name: Controller Method signature: private void handlePrintScheduleByLocation()			
Test Case #	Functional Requirement, or Test Objective	Test description and Input Data	Expected Result
1	This method should print all available sections ordered by location.	- Click Print Schedule by Location	Output displays all initialized sections in classroom/location order.
2	This method should group sections based on classroom/location ordering	- Click Print Schedule by Location	Sections appear ordered by classroom/location rather than by course number.

Java class name: Controller Method signature: private void handlePrintTuitionReport()			
Test Case #	Functional Requirement, or Test Objective	Test description and Input Data	Expected Result
1	This method should print a tuition report for a resident student	- Add a resident student \n- Enroll the student in one or more sections \n-	Output displays the student, enrolled credits, and resident tuition amount.

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		Click Print Tuition Report	
2	This method should print a tuition report for a TriState student.	- Add a TriState student \n- Enroll the student in one or more sections \n- Click Print Tuition Report	Output displays the student, enrolled credits, and tuition reflecting TriState rules
3	This method should print a tuition report for an international student	- Add an international student \n- Enroll the student in one or more sections \n- Click Print Tuition Report	Output displays the student, enrolled credits, and tuition reflecting international fees.
4	This method should show an empty-list message if no students exist	- Start program with no students added \n- Click Print Tuition Report	Output displays: Student list is empty!

Java class name: Controller Method signature: private void handleClearOutput()			
Test Case #	Functional Requirement, or Test Objective	Test description and Input Data	Expected Result
1	This method should clear the output area	- Generate output using any report or action \n- Click Clear Output	Output area becomes empty.